

Zoning, Transportation Investment, and Urban Sprawl

Research Proposal Presentation

Ke Jiang

Southern Methodist University

April 2026

Zoning and city structure

- Restrictive zoning limits housing supply in desirable area, and forms job-housing separation.
- Households may be pushed toward cheaper but more distant locations.
- This raises commuting costs and sprawling development.

Transportation and city structure

- Rail transit can concentrate development around connected corridors and stations.
- Highway expansion often facilitates decentralization and suburbanization.

Key idea

The effect of transportation infrastructure depends on whether land around accessible locations is allowed to densify.

Research Question

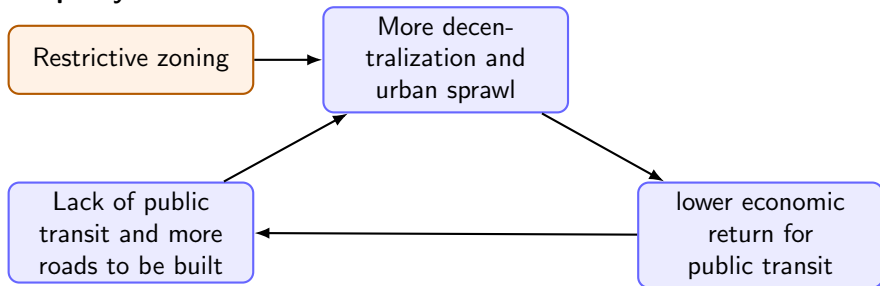
- Restrictive zoning encourage sprawling development
- economic return to public transit is lower in low-density city(Severen, 2021), investment turns towards roadways.
- Lack of public transit cause lower density development
- self-reinforcing...

Central Question

Does the zoning restriction actually have a larger impact if considering its effect on endogeneous public transit investment decisions?

Research Question

For policymakers:



Relation to the Literature

- **Zoning and housing supply:** Glaeser and Kahn (2003), Hsieh and Moretti (2019), Baum-Snow and Han (2024), Rollet and Weiwu (2025), Glaeser and Gyourko (2002), Bordeu (2026), Kashner and Ross (2025)
- **Transportation and congestion externality:** Severen (2021), Allen and Arkolakis (2025), Baum-Snow (2007), Fajgelbaum et al. (2023)
- **Urban sprawl:** Brueckner (2000), Glaeser and Kahn (2003), Coeurdacier, Oswald, and Teignier (2022)

Main Contribution

- 1 I want to show that the cost and benefit of public transit investment is in fact not exogenous but endogenous to zoning regulations.
- 2 How much is the welfare loss by comparing the counterfactual where the zoning regulation is endogenized.

Empirical facts(ongoing)

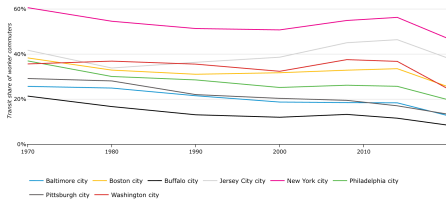
Planned works

- 1 Transportation investment decisions are driven by the economic returns (cost and benefits)
- 2 "Old" cities with developed metros before car-dominance have high returns and more transit, whereas "new" cities are opposite
- 3 Zoning laws lead to lower return to transit investment.

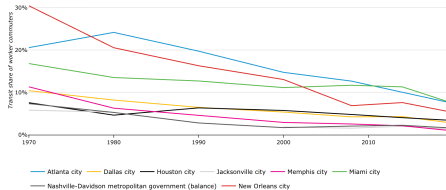
Data source:

- National Transit Database(NTD), Transit Explore Database: "Transit volumes, revenues and investment, operating date"
- Zoning: housing supply elasticity from Baum-Snow and Han (2024)

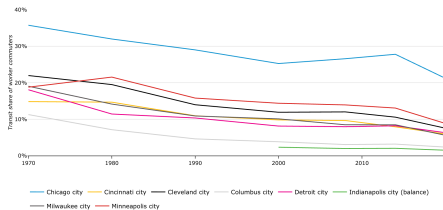
Some Facts



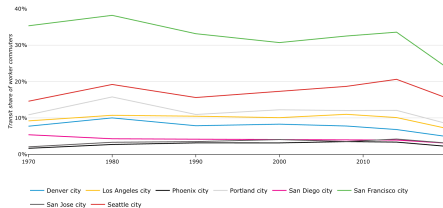
Northeast



Southeast



Midwest



West

Model Environment

Consider a metropolitan area with locations $S = \{1, 2, \dots, N\}$ partitioned into municipalities $g \in G$.

- Each location has fixed land endowment $\bar{H}_i$.
- Land can be allocated between residential and commercial use:

$$H_i^R + H_i^F \leq \bar{H}_i.$$

- Zoning restricts feasible development intensity in each location.
- Commuting costs depend on transportation infrastructure and congestion (Bordeu, 2026):

$$\tau_{ijr} = \tau_{ijr}(I^R, I^T, n).$$

There are three types of agents: households, firms, and governments.

Households and Firms

Households

- Choose residence location i
- Choose workplace location j
- Choose commuting route $r \in R_{ij}$
- Value wages, rents, amenities, and commuting cost

A convenient reduced-form indirect utility is:

$$V_{ijr}(\omega) \propto \frac{B_i}{\tau_{ijr}} \frac{w_j}{(q_i^R)^{1-\beta}} v_{ijr}(\omega).$$

Household sorting across space depends on wages, rents, amenities, and infrastructure-induced commuting costs.

Firms

- Produce a tradable good at workplace location j
- Use labor L_j and commercial land H_j^F

$$Y_j = A_j F(L_j, H_j^F).$$

Competition determines labor demand and commercial land demand.

Government Structure

Municipal governments

- Choose local zoning policies Z_g
- Act as reduced-form representatives of incumbent residents
- Care about local property values, amenities, and congestion/crowding

$$W^g = \sum_{i \in S^g} [\omega_1 P_i + \omega_2 B_i^{\text{loc}} - \omega_3 N_i],$$

Metropolitan authority

- Chooses road and transit investment
- Internalizes regional transport benefits
- Takes local zoning as given

$$\max W^M(I^R, I^T; Z) - C(I^R, I^T),$$

Two-Period Mechanism

Timing

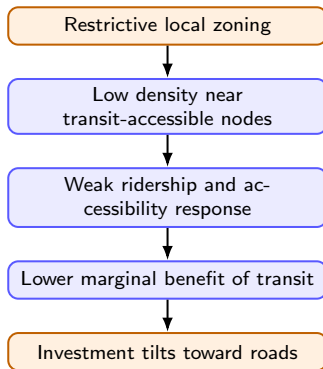
- 1 Municipalities choose zoning capacity $h_{0,i}$ to protect incumbent interests.
- 2 The transportation authority observes h_0 and chooses road/transit investment.
- 3 Households sort across residences, jobs, and commuting modes.

Decentralized wedge

Local zoning omits two metropolitan benefits:

- lower rents for households outside the municipality,
- higher returns to transit investment near dense nodes.

$$h_{0,i}^{DE} < h_{0,i}^{SP}$$



Key implication: zoning changes the relative return of transit versus road investment.

Efficient Benchmark

Planner's problem

A coordinated planner jointly chooses transportation investment and zoning to maximize aggregate metropolitan welfare.

$$\max_{I^R, I^T, Z} W(I^R, I^T, Z) - C(I^R, I^T),$$

Comparing the decentralized equilibrium to this benchmark reveals whether fragmented governance leads to excessive sprawl and underutilized transit-oriented development.

Reference

-  Allen, Treb and Costas Arkolakis (2025). “Quantitative regional economics”. en. In: *Handbook of Regional and Urban Economics*. Vol. 6. Handbook of regional and urban economics. Elsevier, pp. 1–72.
-  Baum-Snow, Nathaniel (2007). “Did Highways Cause Suburbanization?” In: *The Quarterly Journal of Economics* 122 (2), pp. 775–805.
-  Baum-Snow, Nathaniel and Lu Han (June 1, 2024). “The microgeography of housing supply”. en. In: *The Journal of Political Economy* 132 (6), pp. 1897–1946.
-  Bordeu, Olivia (2026). *Commuting Infrastructure in Fragmented Cities*.
-  Brueckner, Jan K (Apr. 2000). “Urban Sprawl: Diagnosis and Remedies”. en. In: *International Regional Science Review* 23 (2), pp. 160–171.
-  Coeurdacier, Nicolas, Florian Oswald, and Marc Teignier (2022). “Structural Change, Land Use, and Urban Expansion”. In: *SSRN Electronic Journal*.
-  Fajgelbaum, Pablo et al. (July 1, 2023). *Political preferences and transport infrastructure: Evidence from California’s high-speed rail*. Research rep. Cambridge, MA.
-  Glaeser, Edward L and Joseph Gyourko (2002). *The Impact of Zoning on Housing Affordability*. en. Research rep. 8835. Cambridge, MA.
-  Glaeser, Edward L and Matthew E Kahn (2003). *Sprawl and Urban Growth*.^{13 / 14}